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Method and device for audio input routing

Abstract

A method on a mobile device for a wireless network is described. An audio input is monitored for a trigger phrase spoken by a user of the mobile device. A command phrase spoken by the user after the trigger phrase is buffered. The command phrase corresponds to a call command and a call parameter. A set of target contacts associated with the mobile device is selected based on respective voice validation scores and respective contact confidence scores. The respective voice validation scores are based on the call parameter. The respective contact confidence scores are based on a user context associated with the user. A call to a priority contact of the set of target contacts is automatically placed if the voice validation score of the priority contact meets a validation threshold and the contact confidence score of the priority contact meets a confidence threshold.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to processing audio signals and, more particularly, to methods and devices for audio signals including voice or speech.

BACKGROUND

(2) Although speech recognition has been around for decades, the quality of speech recognition software and hardware has only recently reached a high enough level to appeal to a large number of consumers. One area in which speech recognition has become very popular in recent years is the smartphone and tablet computer industry. Using a speech recognition-enabled device, a consumer can perform such tasks as making phone calls, writing emails, and navigating with GPS, strictly by voice.

(3) Speech recognition in such devices is far from perfect, however. When using a speech recognition-enabled device for the first time, the user may need to “train” the speech recognition software to recognize his or her voice. Even after training, however, the speech recognition functions may not work well in all sound environments. An unintended call that is placed may result from inaccurate voice recognition of audio that is captured. Occurrences of these types of calls could result in negative financial, legal and social implications, and may lead to user avoiding the use of a voice recognition system for making calls.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) While the appended claims set forth the features of the present techniques with particularity, these techniques, together with their objects and advantages, may be best understood from the following detailed description taken in conjunction with the accompanying drawings of which:

(2) FIG. 1 is a block diagram illustrating a mobile device, according to an embodiment;

(3) FIG. 2 is a block diagram of example components of a mobile device, according to an embodiment;

(4) FIG. 3, FIG. 4, and FIG. 5 illustrate a process flow of a method for validating a command phrase that may be performed by the mobile device of FIG. 1, according to an embodiment;

(5) FIG. 6 is a process flow of a method for automatically dialing a priority contact that may be performed by the mobile device of FIG. 1, according to an embodiment;

(6) FIG. 7 is a process flow of a method for automatically dialing a phone number that may be performed by the mobile device of FIG. 1, according to an embodiment.

DETAILED DESCRIPTION

(7) Turning to the drawings, wherein like reference numerals refer to like elements, techniques of the present disclosure are illustrated as being implemented in a suitable environment. The following description is based on embodiments of the claims and should not be taken as limiting the claims with regard to alternative embodiments that are not explicitly described herein.

(8) Avoidance of placing of unintended calls (e.g., “ghost calls”) is particularly relevant for mobile devices that provide an “always on” voice recognition system. In an always-on system, the mobile device may be configured to monitor or listen for a trigger phrase spoken by a user of the mobile device, followed by a command phrase. Trigger phrases spoken by the user, such as “OK, Google Now” may be previously “learned” in a training session for the mobile device; however, the trigger phrase may be incorrectly identified due to various factors, such as background noise or obstructions between the user and the mobile device.

(9) Validation of the command phrase allows for a reduction in the placement of unintended calls.

The command phrase for placing a call includes a call command and a call parameter. As one example, the mobile device may detect “OK Google Now, Call Bob on his cell phone.” In this case, “OK Google Now” is the trigger phrase, “Call” is the call command, and “Bob on his cell phone” is the call parameter. As another example, the mobile device may detect “OK Google Now, Call Charlie.” In this case, the trigger phrase and call command are the same as the previous example, but the call parameter is simply “Charlie.” As a third example, the mobile device may detect “OK Google Now, Dial 1-800-555-1234.” In this case, “OK Google Now” is the trigger phrase, “Dial” is the call command, and “1-800-555-1234” is the call parameter.

(10) The various embodiments described herein allow a mobile device to validate the command phrase for placing a call. For example, if the call parameter is readily recognized as “Charlie”, the mobile device may validate the command phrase by searching an address book. If the address book includes a frequently called contact with a name of “Charlie”, the mobile device may automatically place a call to that contact without further action by the user. Alternatively, if the call parameter is recognized as spoken digits such as “1-800-555-1234”, the mobile device may validate the digits against a phone numbering plan prior to placing the call.

(11) In one embodiment, a set of target contacts are selected based on respective voice validation scores and respective contact confidence scores. A phone call is automatically placed to a priority contact if a voice validation score and contact confidence score for the priority contact meet a validation threshold.

(12) In another embodiment, numeric digits detected in a command phrase are validated as a phone number against a predetermined numbering plan. A call is automatically placed to the phone number if the phone number is validated.

(13) Referring to FIG. 1, there is illustrated a perspective view of an example mobile device **100**). The mobile device **100** may be any type of device capable of storing and executing multiple applications. Examples of the mobile device **100** include, but are not limited to, mobile devices, smart phones, smart watches, wireless devices, tablet computing devices, personal digital assistants, personal navigation devices, touch screen input device, touch or pen-based input devices, portable video and/or audio players, and the like. It is to be understood that the mobile device **100** may take the form of a variety of form factors, such as, but not limited to, bar, tablet, flip/clam, slider, rotator, and wearable form factors.

(14) For one embodiment, the mobile device **100** has a housing **101** comprising a front surface **103** which includes a visible display **105** and a user interface. For example, the user interface may be a touch screen including a touch-sensitive surface that overlays the display **105**. For another embodiment, the user interface or touch screen of the mobile device **100** may include a touch-sensitive surface supported by the housing **101** that does not overlay any type of display. For yet another embodiment, the user interface of the mobile device **100** may include one or more input keys **107**. Examples of the input key or keys **107** include, but are not limited to, keys of an alpha or numeric keypad or keyboard, a physical keys, touch-sensitive surfaces, mechanical surfaces, multipoint directional keys and side buttons or keys **107**. The mobile device **100** may also comprise a speaker **109** and microphone **111** for audio output: and input at the surface. It is to be understood that the mobile device **100** may include a variety of different combination of displays and interfaces.

(15) The mobile device **100** includes one or more sensors **113** positioned at or within an exterior boundary of the housing **101**. For example, as illustrated by FIG. 1, the sensor or sensors **113** may be positioned at the front surface **103** and/or another surface (such as one or more side surfaces **115**) of the exterior boundary of the housing **101**. The sensor or sensors **113** may include an exterior sensor supported at the exterior boundary to detect an environmental condition associated with an environment external to the housing. The sensor or sensors **113** may also, or in the alternative, include an interior sensors supported within the exterior boundary (i.e., internal to the housing) to detect a condition of the device itself. Examples of the sensors **113** are described below

in reference to FIG. 2.

(16) Referring to FIG. 2, a block diagram representing example components **200** that may be used for one or more embodiments is shown. The example components may include one or more communication components **201**, one or more processors **203**, one or more memories **205**, one or more output components **207**, and one or more input components **209**. Each component may include a user interface that comprises one or more input components **209**. Each communication component **201** may include a wireless receiver, transmitter or transceiver. Each communication component **201** may utilize wireless technology for communication, such as, but are not limited to, cellular-based communications such as analog communications (using AMPS), digital communications (using CDMA, TDMA, GSM, iDEN, GPRS, or EDGE), and next generation communications (using UMTS, WCDMA, LTE, or IEEE 802.16) and their variants, as represented by cellular transceiver **211**. Each communication component **201** may also utilize wireless technology for communication, such as, but are not limited to, peer-to-peer or ad hoc communications such as HomeRF, Bluetooth and IEEE 802.11 (a, b, g or n); and other forms of wireless communication such as infrared technology, as represented by WLAN transceiver **213**. Also, each communication component **201** may be a receiver, a transmitter or both.

(17) The example components **200** may further include a device interface **215** to provide a direct connection to auxiliary components or accessories for additional or enhanced functionality. In addition, the example components **200** may include a power source or supply **217**, such as a portable battery, for providing power to the other example components and allow portability of the mobile device **100**.

(18) The processor **203** may generate commands based on information received from one or more communication components **201** and/or one or more input components **209**. The processor **203** may process the received information alone or in combination with other data, such as the information stored in the memory **205**. Thus, the memory **205** of the example components **200** may be used by the processor **203** to store and retrieve data. The data that may be stored by the memory **205** include, but is not limited to, operating systems, applications, and data. Each operating system includes executable code that controls basic functions of the mobile device, such as interaction among the components of the example components **200**, communication with external devices via each communication component **201** and/or the device interface (see below), and storage and retrieval of applications and data to and from the memory **205**. The memory **205** includes multiple applications, and each application includes executable code utilizes an operating system to provide more specific functionality for the mobile device. Data is non-executable code or information that may be referenced and/or manipulated by an operating system or application for performing functions of the mobile device.

(19) The input components **209**, such as components of the user interface, may produce an input signal in response to detecting a predetermined gesture at a first input component **219**, such as a gesture sensor. An example of a gesture sensor is, but not limited to, a touch-sensitive sensor having a touch-sensitive surface substantially parallel to the display. The touch-sensitive sensor may include at least one of a capacitive touch sensor, a resistive touch sensor, an acoustic sensor, an ultrasonic sensor, a proximity sensor, or an optical sensor.

(20) The input components **209** may also include other sensors, such as the visible light sensor, the motion sensor and the proximity sensor described above. Likewise, the output components **207** of the example components **200** may include one or more video, audio and/or mechanical outputs. For example, the output components **207** may include a video output component such as a cathode ray tube, liquid crystal display, plasma display, incandescent light, fluorescent light, front or rear projection display, and light emitting diode indicator. Other examples of output components **207** include an audio output component such as a speaker, alarm and/or buzzer, and/or a mechanical output component such as vibrating or motion-based mechanisms.

(21) It is to be understood that FIG. 2 is provided for illustrative purposes only and for illustrating

components of a mobile device in accordance with the present invention, and is not intended to be a complete schematic diagram of the various components required for a mobile device. Therefore, a mobile device may include various other components not shown in FIG. 2, or may include a combination of two or more components or a division of a particular component into two or more separate components, and still be within the scope of the present invention.

(22) Turning to FIGS. 3, 4, and 5, process flows **300**, **400**, and **500** illustrate an embodiment of a method for validation of a command phrase that may be performed by the mobile device **100**. As a user of the mobile device **100** is speaking, the mobile device **100** monitors an audio input, such as an audio input from the microphone **111**. Upon detection of a trigger phrase (e.g., “OK Google Now”), the mobile device **100** stores a command phrase that follows the trigger phrase. The command phrase or a portion thereof in one example is buffered in the memory **205** and provided to the processor **203**. In another example, the command phrase is provided directly to the processor **203**. The mobile device **100** determines whether the command phrase includes a call command for a current call. Examples of a call command include “Call”, “Dial”, “Place a call to”, and other words or phrases that will be apparent to those skilled in the art. The call commands in one example are included in a call command whitelist.

(23) When the mobile device **100** recognizes (302) a call command, the mobile device **100** determines a command confidence score for the call command. In one example, the mobile device **100** analyzes the command phrase or a portion thereof (e.g., only the call command) with a voice recognition engine, such as those provided by Nuance (Nuance Communications, Inc; Burlington, MA), or other processing algorithm to determine the command confidence score. The command confidence score in this case is a numerical value (e.g., between 0 and 100 with 100 being most confident) that indicates a relative confidence level of the detection of a call command. The mobile device **100** in one example provides the call command whitelist and the call command to the voice recognition engine to receive the command confidence score. Alternatively, the call command whitelist may be provided to the voice recognition engine prior to the call command (e.g., during installation, configuration, initialization, etc.). The mobile device **100** determines (304) whether the command confidence score meets a command threshold (e.g., is greater than or equal to 40). In alternative embodiments, the command confidence score is determined or received as a valid or invalid indication, or a tiered response indication (e.g., high confidence, medium confidence, low confidence) Other implementations of the command confidence score will be apparent to those skilled in the art.

(24) In addition to determining (304) the command confidence score, the mobile device **100** may also compare the call command to a list of predetermined words or phrases that may be misidentified as call commands. For example, the mobile device **100** may use a blacklist of words or phrases such as “How tall is . . .” or “denial” which may be misidentified as “Call” or “Dial”, respectively. If the command phrase is a closer match to the blacklist phrase than to a legitimate call command (e.g., on the whitelist), then the mobile device **100** proceeds to step **306**.

Additionally, the mobile device **100** may determine whether the command phrase is located in the middle of a conversation. In this case, the mobile device **100** determines whether an audio length of the command phrase is longer than a predetermined threshold, such as five seconds, to prevent an attempt to validate a user's conversation. The mobile device **100** invalidates the call command if the audio length exceeds the predetermined threshold and proceeds to step **306**.

(25) If the command threshold is not met (NO at **304**), the mobile device **100** analyzes (306) the command phrase with a voice command engine (e.g., engines provided by Google Now, Sensory, or Nuance), for example, to provide general command recognition instead of call command validation. In this case, the command confidence score indicates that the call command may have been incorrectly identified. For example, if the command phrase is “How tall is Mount Everest?”, “low tall is” may be recognized as a blacklist phrase, but additional processing by the voice command engine may allow a response to be provided.

(26) If the command threshold is met (YES at **304**), the mobile device **100** determines (**308**) whether the call parameter includes spoken digits (e.g., “1-800-555-1234”). If spoken digits are detected (YES at **308**), the mobile device **100** validates (**310**) the spoken digits (e.g., the call parameter) against a predetermined phone numbering plan. Examples of phone numbering plans include the North American Numbering Plan, E.164, satellite phone number plans, or private numbering plans (e.g., for a private branch exchange). The mobile device **100** in one example selects a numbering plan based on its current geographic location (e.g., North America or Europe). In another example, the numbering plan includes a blacklist of phone numbers. The blacklist includes phone numbers that should not be automatically dialed, such as emergency response numbers (e.g., 9-1-1), international phone numbers, or phone numbers that may incur additional fees. The blacklist may be populated by a wireless service provider, government entity, or by the user of the mobile device **100**. If the call parameter is not validated (NO at **310**), the mobile device **100** proceeds to step **306**. If the call parameter is validated and thus contains a phone number (YES at **310**), the mobile device **100** proceeds to step **504** (FIG. 5). Optionally, the mobile device **100** determines a voice validation score for the phone number that indicates a relative confidence level of the detection of phone number (e.g., to avoid mistaken detections such as detecting a “live” as a “nine”). The mobile device **100** uses the voice recognition engine for determination of the voice validation score.

(27) If spoken digits are not detected (NO at **308**), the mobile device **100** validates the call parameter against a set of call contacts, such as an address book associated with the mobile device **100**. The mobile device **100** generates (**312**) a set of voice validation scores that correspond to the set of call contacts in the address book. The voice validation scores indicate a relative confidence level of the detection of the corresponding contact's spoken name. In one example, the mobile device **100** provides the set of call contacts to the voice recognition engine along with the call parameter. The set of call contacts may be provided to the voice recognition engine prior to the call parameter (e.g., during installation, configuration, initialization, etc.). The set of voice validation scores in one example are numerical values (e.g., 0 to 100). In alternative embodiments, the voice validation score received as a valid or invalid indication, or a tiered response indication (e.g., high confidence, medium confidence, low confidence). Other implementations of the voice validation score will be apparent to those skilled in the art.

(28) The mobile device **100** selects (**314**) a set of target contacts from the set of call contacts based on the voice validation scores. For example, the mobile device **100** selects only those contacts with a voice validation score within a selected range of a highest voice validation score. As one example, the user may have call contacts in his address book including Andrea, Brad, Brent, and Charlie. In this case, if the user speaks “OK Google Now, Call Brent at his home”, the voice validation scores may be 0, 85, 95, and 0, respectively. If the selected range is 10, the target contacts are Brad and Brent (Brent having the highest voice validation score of 95). If the selected range is 5, the target contact is only Brent. The selected range may be predetermined or dynamically determined by the mobile device **100** (e.g., based on prior voice recognition history). In alternative implementations, the selected range may include “high confidence” voice validation scores or both “high confidence” and “medium confidence” scores. Other selected ranges will be apparent to those skilled in the art, for example, based on the type of voice validation score used.

(29) Once the set of target contacts have been selected (e.g., those with a sufficiently high voice validation score), the mobile device **100** generates a contact confidence score for the target contacts. The contact confidence scores indicate a relative importance of the corresponding contact to the user of the mobile device **100** and may be a numbered value (e.g., 0 to 100), a ranked list, or other relative indicator for comparison of the contacts. The mobile device **100** in one example generates the contact confidence scores for each of the set of target contacts based on a user context associated with the user. The user context of a contact includes one or more of a recency of last communication (e.g., phone calls, text messages, emails, etc.), frequency of communication,

placement on a “favorites list” by the user, relation to a calendar event or appointment (e.g., a current calendar event of “Call Brent at 10:00 AM”), or other contextual indicators as will be apparent to those skilled in the art.

(30) In an alternative implementation, the mobile device **100** generates the contact confidence scores prior to recognizing (312) the call command. For example, the mobile device **100** may use a context engine that analyzes the user context to update contact confidence scores on a schedule-driven basis (e.g., a predetermined schedule or dynamic schedule) or on an event-driven basis (e.g., when a communication is sent to a contact). In this case, the mobile device **100** (e.g., the context engine) may maintain a “VIP” list of contacts with their respective contact confidence scores. The context engine in one example is an application or service that is executed by the mobile device **100**.

(31) The mobile device **100** selects (318) respective phone numbers for the set of target contacts. As one example where the call parameter is “Brent at his home”, the mobile device **100** detects a phone type specified within the call parameter, such as “at his home”. The mobile device **100** in one example uses a whitelist of words or phrases that specify phone types, such as for a cell or mobile phone, home phone, business phone, or others as will be apparent to those skilled in the art. As another example, the call parameter may not include a phone type and simply identify a contact (e.g., “Call Brent”).

(32) The mobile device **100** selects a most likely phone number based on the specified phone type (if available) and the user context. In one example, the mobile device **100** queries the context engine with the specified phone type to receive the most likely phone number. If the specified phone type is not available or if no phone number is received from a query with the specified phone type (e.g., “at his home” but no home phone number is listed in the contact), the mobile device **100** queries without the phone type specified. In this case, the context engine may provide a most likely number based on user context or selected from a predetermined priority order as available (e.g., mobile phone as highest priority, then home, then work). Where the user context is not available for a contact or if the contact is not on the VIP list, the mobile device **100** selects the phone number based on the predetermined priority order.

(33) Referring to FIG. 4, after selection (318) of the phone numbers, the mobile device **100** determines (402) whether the set of target contacts contains only a single contact without a specified phone type. In this case (YES at 402), the mobile device **100** obtains each of the phone numbers stored (e.g., in the address book) for the single contact and proceeds to step 408. If the set of target contacts contains more than one contact or has a specified phone type, the mobile device **100** determines (406) whether a priority contact for the current call has been found, based on the voice validation score, the contact confidence score, or a combination thereof. In one example, a priority contact is a contact with a voice validation score that meets a validation threshold (e.g., at least 60), a contact confidence score that meets a confidence threshold (e.g., at least 60), or with both thresholds met. In another example, the mobile device **100** uses a weighting algorithm based on the voice validation score and contact confidence score to provide a higher likelihood of “priority” status for a contact on the VIP list. If the validation and confidence thresholds are met and a priority contact is found, the mobile device **100** proceeds to step 502 (FIG. 5).

(34) If a priority contact is not found (NO at 406), or after obtaining (404) the phone numbers, the mobile device **100** sorts the target contacts and phone numbers. The mobile device **100** in one example sorts the contacts and numbers using a weighting algorithm based on the voice validation score and contact confidence score. After sorting, the mobile device **100** proceeds to step 502 (FIG. 5).

(35) Referring to FIG. 5, the mobile device **100** determines (502) whether the set of target contacts contains a single entry (e.g., a single contact with a single selected phone number). If a single entry is not found (NO at 502), the mobile device **100** prompts (506) the user to indicate that multiple entries (e.g., multiple target contacts or multiple phone numbers) have been found. The

communication (**506**) in one example is a notification that multiple entries have been found and a prompt for further input. In another example, the communication is a list of the multiple entries (or a portion thereof).

(36) The mobile device **100** communicates the multiple entries by a visual interface (e.g., on the display **105**), a text-to-speech interface (e.g., via the speaker **109**), or a combination thereof. In one example, the mobile device **100** communicates only a portion of the multiple entries selected based on their relative voice validation scores or contact confidence scores. In a further example, the communication includes a prompt for the user to select one or more user interface options, such as communicating all of the entries found, sequentially listing the entries, or selecting an entry for placement of the current call. If the user selects an option for communicating the entries (YES at **508**), the mobile device **100** communicates (**510**) the found entries (e.g., contacts or phone numbers) via the visual or text-to-speech interface.

(37) The mobile device **100** determines (**512**) whether an entry has been selected by the user. Upon selection of an entry (YES at **512**), the mobile device **100** places (**514**) a call to the selected entry. If the user does not select an entry within a predetermined timeout period such as five seconds (NO at **512**), the process ends for the current call.

(38) If a single entry is found (YES at **502**), or if a valid phone number is found (YES at **310**), the mobile device **100** determines (**504**) whether the single entry is a priority contact based on the voice validation score, the contact confidence score, or a combination thereof. As described above, a priority contact may be a contact with a voice validation score that meets the validation threshold or a contact confidence score that meets the contact threshold. The mobile device **100** may determine that a valid phone number (YES at **310**) is a priority contact if the corresponding voice validation score meets the validation threshold. However, other thresholds may be used at step **504** for determination of the priority contact. In one example, the mobile device **100** dynamically determines at least one of the validation threshold or contact threshold based on a device context for the mobile device **100**. In this case, the mobile device **100** selects a higher threshold when the mobile device **100** is currently stowed (e.g., in a pocket or bag), the user is currently driving, the user is currently in a meeting, whether a headset (e.g., Bluetooth or wired headset) is currently connected, or other criteria, as will be apparent to those skilled in the art.

(39) Upon determination that a priority contact has been found (YES at **504**), the mobile device **100** communicates (**516**) that an automatic call placement is (or will) take place, for example, via the visual or text-to-speech interface (e.g., “Calling Brent at home”). This provides an opportunity for the user to cancel the call prior to automatic placement of the call. The mobile device **100** determines (**518**) whether the user has canceled the call, for example, within a predetermined timeout period (e.g., four seconds). If the user cancels the call (YES at **518**), for example, via a spoken command (“Cancel”) or other interaction with the mobile device **100**, the process ends for the current call. If the user does not cancel the call (NO at **518**), the mobile device **100** automatically places the call to the priority contact.

(40) If a priority contact has not been found (NO at **504**), the mobile device **100** prompts (**520**) the user with a call confirmation via the visual or text-to-speech interface (e.g., “Should I call Brent at home?” or “Press here to place call”). The mobile device **100** determines (**522**) whether the user has confirmed the call, for example, within a predetermined timeout period (e.g., ten seconds). If the user confirms the call (YES at **522**), for example, via a spoken command (“Yes”) or other interaction with the mobile device **100**, the mobile device **100** places (**514**) the call. If the user does not confirm the call (NO at **522**), the process ends for the current call.

(41) Turning to FIG. **6**, a process flow **600** for automatically dialing a priority contact that may be performed by the mobile device **100** is shown. The mobile device **100** monitors (**602**) an audio input (e.g., the microphone **111**) for a trigger phrase spoken by a user of the mobile device **100**. In one example, the mobile device **100** performs the monitoring in a low power mode, such as with a low power digital signal processor while an application processor is in a low power state. The

mobile device **100** buffers (**604**) a command phrase spoken by the user after the trigger phrase (**604**). The command phrase corresponds to a call command and a call parameter. The mobile device **100** in one example wakes from the low power mode (e.g., wakes the application processor) upon detection of the trigger phrase and buffers the command phrase while the application processor is initialized. The mobile device **100** selects (**606**) a set of target contacts associated with the mobile device based on respective voice validation scores and respective contact confidence scores. The respective voice validation scores are based on the call parameter and the respective contact confidence scores are based on a user context associated with the user. The mobile device **100** determines (**608**) whether the voice validation score and contact confidence score for a priority contact meet corresponding validation and contact thresholds. The mobile device **100** automatically places (**610**) the call to the priority contact if the thresholds are met (YES at **608**).

(42) Turning to FIG. 7, a process flow **700** for automatically dialing a phone number that may be performed by the mobile device **100** is shown. The mobile device **100** monitors (**702**) an audio input (e.g., the microphone **111**) for a trigger phrase spoken by a user of the mobile device **100**. In one example, the mobile device **100** performs the monitoring in a low power mode, such as with a low power digital signal processor while an application processor is in a low power state. The mobile device **100** buffers (**704**) a command phrase spoken by the user after the trigger phrase (**604**). The command phrase corresponds to a call command and a call parameter. The mobile device **100** in one example wakes from the low power mode (e.g., wakes the application processor) upon detection of the trigger phrase and buffers the command phrase while the application processor is initialized. If numeric digits are detected in the call parameter, the mobile device **100** validates (**706**) the numeric digits as a phone number against a predetermined phone numbering plan. If the digits are validated (YES at **708**), the mobile device **100** automatically places (**710**) a call to the phone number.

(43) It can be seen from the foregoing that a method and system for validation of a command phrase for placing a call have been described. In view of the many possible embodiments to which the principles of the present discussion may be applied, it should be recognized that the embodiments described herein with respect to the drawing figures are meant to be illustrative only and should not be taken as limiting the scope of the claims. Therefore, the techniques as described herein contemplate all such embodiments as may come within the scope of the following claims and equivalents thereof.

(44) The apparatus described herein may include a processor, a memory for storing program data to be executed by the processor, a permanent storage such as a disk drive, a communications port for handling communications with external devices, and user interface devices, including a display, touch panel, keys, buttons, etc. When software modules are involved, these software modules may be stored as program instructions or computer readable code executable by the processor on a non-transitory computer-readable media such as magnetic storage media (e.g., magnetic tapes, hard disks, floppy disks), optical recording media (e.g., CD-ROMs, Digital Versatile Discs (DVDs), etc.), and solid state memory (e.g., random-access memory (RAM), read-only memory (ROM), static random-access memory (SRAM), electrically erasable programmable read-only memory (EEPROM), flash memory, thumb drives, etc.). The computer readable recording media may also be distributed over network coupled computer systems so that the computer readable code is stored and executed in a distributed fashion. This computer readable recording media may be read by the computer, stored in the memory, and executed by the processor.

(45) The disclosed embodiments may be described in terms of functional block components and various processing steps. Such functional blocks may be realized by any number of hardware and/or software components configured to perform the specified functions. For example, the disclosed embodiments may employ various integrated circuit components, e.g., memory elements, processing elements, logic elements, look-up tables, and the like, which may carry out a variety of functions under the control of one or more microprocessors or other control devices. Similarly,

where the elements of the disclosed embodiments are implemented using software programming or software elements, the disclosed embodiments may be implemented with any programming or scripting language such as C, C++, JAVA®, assembler, or the like, with the various algorithms being implemented with any combination of data structures, objects, processes, routines or other programming elements. Functional aspects may be implemented in algorithms that execute on one or more processors. Furthermore, the disclosed embodiments may employ any number of conventional techniques for electronics configuration, signal processing and/or control, data processing and the like. Finally, the steps of all methods described herein may be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context.

(46) For the sake of brevity, conventional electronics, control systems, software development and other functional aspects of the systems (and components of the individual operating components of the systems) may not be described in detail. Furthermore, the connecting lines, or connectors shown in the various figures presented are intended to represent exemplary functional relationships and/or physical or logical couplings between the various elements. It should be noted that many alternative or additional functional relationships, physical connections or logical connections may be present in a practical device. The words “mechanism”, “element”, “unit”, “structure”, “means” “device”, “controller”, and “construction” are used broadly and are not limited to mechanical or physical embodiments, but may include software routines in conjunction with processors, etc.

(47) No item or component is essential to the practice of the disclosed embodiments unless the element is specifically described as “essential” or “critical”. It will also be recognized that the terms “comprises,” “comprising,” “includes,” “including,” “has,” and “having,” as used herein, are specifically intended to be read as open-ended terms of art. The use of the terms “a” and “an” and “the” and similar referents in the context of describing the disclosed embodiments (especially in the context of the following claims) are to be construed to cover both the singular and the plural, unless the context clearly indicates otherwise. In addition, it should be understood that although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms, which are only used to distinguish one element from another.

Furthermore, recitation of ranges of values herein are merely intended to serve as a shorthand method of referring individually to each separate value falling within the range, unless otherwise indicated herein, and each separate value is incorporated into the specification as if it were individually recited herein.

(48) The use of any and all examples, or exemplary language (e.g., “such as”) provided herein, is intended merely to better illuminate the disclosed embodiments and does not pose a limitation on the scope of the disclosed embodiments unless otherwise claimed. Numerous modifications and adaptations will be readily apparent to those of ordinary skill in this art.

Claims

1. A method implemented by one or more processors of a device, the method comprising: monitoring an audio input for an occurrence of a trigger phrase that is spoken by a user of the device while the device is in a low power mode; and in response to detecting the occurrence of the trigger phrase: waking the device from the low power mode; and buffering the audio input while the device wakes up from the low power mode; and in response to the device waking up from the low power mode, processing the buffered audio input to detect an occurrence of a command phrase that is spoken by the user in a portion of the buffered audio input; in response to detecting the occurrence of the command phrase that is spoken by the user in the portion of the buffered audio input, determining whether the command phrase spoken by the user of the device is directed to the device or is located in the middle of a conversation, wherein determining whether the command phrase spoken by the user of the device is directed to the device or is located in the middle of a conversation comprises: determining whether a length of the buffered audio input satisfies a

threshold length; determining, based on the length of the buffered audio input satisfying the threshold length, that the command phrase spoken by the user is located in the middle of the conversation; and in response to determining the command phrase spoken by the user is located in the middle of the conversation and based on the length of the buffered audio input satisfying the threshold length, invalidating the command phrase.

2. The method of claim 1, wherein the command phrase corresponds to a call command and a call parameter.

3. The method of claim 2, wherein invalidating the command phrase comprises: refraining from acting upon the call command using the call parameter.

4. The method of claim 1, wherein the command phrase corresponds to a request for information.

5. The method of claim 1, wherein the threshold length is a predetermined threshold length.

6. A device comprising: at least one processor; and memory storing instructions that, when executed, cause the at least one processor of the device to: monitor an audio input for an occurrence of a trigger phrase that is spoken by a user of the device while the device is in a low power mode; and in response to detecting the occurrence of the trigger phrase: wake the device from the low power mode; and buffer the audio input while the device wakes up from the low power mode; and in response to the device waking up from the low power mode, process the buffered audio input to detect an occurrence of a command phrase that is spoken by the user in a portion of the buffered audio input; in response to detecting the occurrence of the command phrase that is spoken by the user in the portion of the buffered audio input, determine whether the command phrase spoken by the user of the device is directed to the device or is located in the middle of a conversation, wherein the instructions to determine whether the command phrase spoken by the user of the device is directed to the device or is located in the middle of a conversation comprise instructions to: determine whether a length of the buffered audio input satisfies a threshold length; determine, based on the length of the buffered audio input satisfying the threshold length, that the command phrase spoken by the user is located in the middle of the conversation; and in response to determining the command phrase spoken by the user is located in the middle of the conversation and based on the length of the buffered audio input satisfying the threshold length, invalidate the command phrase.

7. The device of claim 6, wherein the command phrase corresponds to a call command and a call parameter.

8. The device of claim 7, wherein the instructions to invalidate the command phrase comprise instructions to: refrain from acting upon the call command using the call parameter.

9. The device of claim 6, wherein the command phrase corresponds to a request for information.

10. The device of claim 6, wherein the threshold length is a predetermined threshold length.

11. A non-transitory computer-readable storage medium storing instructions that, when executed, cause at least one processor of a device to perform operations, the operations comprising: monitoring an audio input for an occurrence of a trigger phrase that is spoken by a user of the device while the device is in a low power mode; and in response to detecting the occurrence of the trigger phrase: waking the device from the low power mode; and buffering the audio input while the device wakes up from the low power mode; and in response to the device waking up from the low power mode, processing the buffered audio input to detect an occurrence of a command phrase that is spoken by the user in a portion of the buffered audio input; in response to detecting the occurrence of the command phrase that is spoken by the user in the portion of the buffered audio input, determining whether the command phrase spoken by the user of the device is directed to the device or is located in the middle of a conversation, wherein determining whether the command phrase spoken by the user of the device is directed to the device or is located in the middle of a conversation comprises: determining whether a length of the buffered audio input satisfies a threshold length; determining, based on the length of the buffered audio input satisfying the threshold length, that the command phrase spoken by the user is located in the middle of the

conversation; and in response to determining the command phrase spoken by the user is located in the middle of the conversation and based on the length of the buffered audio input satisfying the threshold length, invalidating the command phrase.

12. The non-transitory computer-readable storage medium of claim 11, wherein the command phrase corresponds to a call command and a call parameter.

13. The non-transitory computer-readable storage medium of claim 12, wherein invalidating the command phrase comprises: refraining from acting upon the call command using the call parameter.

14. The method of claim 4, wherein invalidating the command phrase comprises: refraining from executing the request for information.

15. The device of claim 9, wherein the instructions to invalidate the command phrase comprise instructions to: refrain from executing the request for information.

16. The non-transitory computer-readable storage medium of claim 11, wherein the command phrase corresponds to a request for information.

17. The non-transitory computer-readable storage medium of claim 16, wherein invalidating the command phrase comprises: refraining from executing the request for information.
